MapleBank Practical Track

Sequential, Practice-First Roadmap · Canadian Banking Data Engineering · 16 Steps

The rule for every step: nothing counts until you've run it in your Databricks workspace and committed an artifact to GitHub. Every step is ~20 minutes and follows the same loop: **Learn** (short concept) → **Lab** (run real code) → **Commit** (artifact to GitHub) → **Verify** (Claude checks your output before advancing).

Your Starting State (Already Done)

Component	Status	Details
Databricks Free Edition	■ Ready	Serverless compute, Python notebooks
Unity Catalog volume	■ Ready	/Volumes/workspace/default/maplebank/ with 4 CSVs
GitHub repo	■ Ready	maplebank-data-platform with folder structure + README

Sample data in your volume: dim_customer.csv (1,000 rows) · dim_account.csv (2,025 rows) · dim_branch.csv (15 rows) · fact_transactions.csv (10,000 rows). All synthetic Canadian banking data — fake SINs, Canadian postal codes, Interac transaction types, CAD amounts.

Phase A — PySpark Hands-On Foundations (Steps 1-4)

Step	Lab You'll Run	GitHub Artifact
1. Load & Explore	Load all 4 CSVs with explicit schemas from your volume; inspect, count, profile nulls	01_load_explore.py
2. Transformations	Filter FINTRAC ≥$10K txns; add HST + flag columns; joins (inner/left/anti); find orphan transactions	02_transformations.py
3. Formats & Partitioning	Convert CSV → Parquet → Delta; partition by date; prove partition pruning with explain()	03_formats_partitioning.py
4. Window Functions & Tuning	Running balance per account; 30-day spend; top-N per branch; broadcast joins; masking UDFs	04_windows_tuning.py

Phase B — Build the Lakehouse (Steps 5-7)

Step	Lab	Artifact
5. Workspace + Utils	Organize /Workspace/MapleBank/; build shared masking-functions notebook; %run it	utils_masking.py
6. Bronze → Silver → Gold	Full pipeline: raw Bronze; PII-masked Silver + quality report; Gold Customer 360 + FINTRAC LCTR	05_bronze_silver_gold.py
7. Delta Lake Deep Dive	Time travel on YOUR tables; MERGE upsert; SCD Type 2 (customer moves Toronto → Mississauga); O	06_delta_lab.py

Phase C — Orchestration with ADF (Steps 8-10)

Step	Lab	Artifact
8. Azure Setup + ADF Concepts	Create Azure free account (Canada Central!); budget alert; ADF instance; ADF building blocks tour	Screenshots in adf/
9. Build the Pipeline	Copy activity + linked services + datasets; debug run	Pipeline JSON
10. Production-ize	Parameterize by business_date; schedule trigger (7 PM ET); monitoring + alerts	Pipeline JSON v2

Honesty note: Free Edition Databricks can't be triggered from ADF — we orchestrate a real storage-to-storage copy, and cover the Databricks-trigger pattern conceptually with full JSON. You'll know exactly what's runnable vs. described, and can say so precisely in interviews.

Phase D — Talend & Banking Theory (Steps 11-12)

Step	What	Artifact
11. Talend ETL Patterns	tMap masking job + reconciliation design (conceptual — free Talend Open Studio was discontinued Jan 2	talend/pii_masking_rules.md + recon design
12. Banking Domain Deep-Dive	OSFI / FINTRAC / PIPEDA; payment rails (Interac, Lynx, ACSS); EOD batch reality — the vocabulary lay	docs/banking_domain_notes.md

Phase E — Capstone & Interview (Steps 13-16)

Step	What	Artifact
13. Fraud Detection Lab	Structuring detection (multiple $9K+ e-Transfers); velocity rules — run on your data	07_fraud_detection.py
14. End-to-End Review + Polish	Architecture diagram; README final pass; screenshots everywhere	Polished repo
15. Resume + 90-Second Pitch	Bullets; project narrative; STAR stories	interview-prep/pitch.md
16. Mock Interview	25 Canadian banking DE questions — graded, then refined	interview-prep/qa.md

How Each Step Works

```
LEARN   (5 min)  — short concept explanation with banking context
LAB     (12 min) — you run provided code in YOUR workspace,
                   using /Volumes/workspace/default/maplebank/ paths
COMMIT  (3 min)  — export notebook → upload to GitHub folder
VERIFY           — paste your output to Claude; advance only when it runs
```

Ground rules

- Strictly sequential — each lab builds on the previous one's output
- All code uses your serverless + Unity Catalog volume setup
- Every step ends with a GitHub commit — by Step 16 the repo tells the whole story
- Stuck? Paste the exact error — debugging is part of the training

The Finish Line

By Step 16 you'll have: a working lakehouse (Bronze/Silver/Gold with PIPEDA masking and FINTRAC reports) running in Databricks, a real ADF pipeline in Azure Canada Central, a fraud detection notebook, a polished public GitHub portfolio, a rehearsed 90-second pitch, and graded answers to the 25 most common Canadian banking data engineering interview questions.

Phase	Steps	Focus
A — PySpark Foundations	1-4	Hands-on transformations, formats, windows
B — Build the Lakehouse	5-7	Medallion architecture + Delta Lake
C — ADF Orchestration	8-10	Azure pipelines, scheduling, monitoring
D — Talend & Domain	11-12	ETL patterns + Canadian banking vocabulary
E — Capstone & Interview	13-16	Fraud lab, polish, pitch, mock interview